Forecasting the Remaining Shelf Life of Individual Apples from Hyperspectral Imaging

William “Chase” Culbertson
https://explodingcb.com | https://sculptorai.org
October 7, 2026

Abstract

Produce shelf life is managed by the batch, even though individual fruit from the same batch can spoil days apart. This project asks whether a single hyperspectral image of an apple can predict how many days remain before that apple spoils, with a calibrated uncertainty range. Per-fruit prediction from spectra has been reported for several fruits, including avocados and kiwifruit, but I found no published work on apples and none that frames the problem as survival analysis, in which fruit still sound at the end of the experiment are kept as censored observations. The proposed rig images about 90 apples once a day with a hyperspectral camera on a motorized cart in a dark, air-conditioned room. Each apple that expires is replaced with a fresh one, for roughly 300 to 400 apples over about 12 weeks. Survival models are trained on the resulting per-apple image series, and the dataset and code will be released.

1. Background and significance

Shelf-life estimates for produce are made at the batch level, and individual fruit vary widely around them. OneThird, a Dutch produce-scanning company, reports that strawberries sold with a standard five-day date last between 3 and 13 days (Fruit Today). With per-fruit estimates, short-lived fruit could be sent to nearby stores or discounted before it spoils. Commercial scanners from OneThird and Apeel already provide such estimates for some fruits, but their models and data are not public.

2. Related work

Table 1 lists the closest studies that predict a fruit's future condition from spectra or images, all on fruits other than apples.

Spectral studies of apples have mostly estimated current quality. They classify storage age (Zhao et al. 2025), predict firmness, sugar content and pH during cold storage (Khaled et al. 2023), or grade how far spoilage has progressed (Wang et al. 2022, 2024). The one forward-looking apple study I found, Ma et al. (2021), predicts the remaining shelf life of Fuji apples from destructive

laboratory measurements. A time series of 11 to 16 decaying apples has been recorded in RGB, but the decay was segmented rather than forecast (Stasenko et al. 2023).

In agriculture, survival analysis of spectral time series has been applied to the decline of strawberry plants in the field (Detka et al. 2026). Self-supervised pretraining on fruit hyperspectral images appears to have been used only for ripeness classification (Varga, Frank and Zell 2023).

Table 1. Closest prior work on per-fruit prediction of future condition.

Work	Fruit and data	Target	Result
Han et al. 2023	476 avocados, one hyperspectral scan each	Days until ripe	RMSE 1.13 to 1.20 days
Davur et al. 2023	80 avocados, imaged daily	Days until ripe	1.17 days mean error
Waseem et al. 2026 (RipeTrack)	Pears, avocados, mangoes, bananas and nectarines, imaged daily by phone (RGB and NIR)	Percent of lifetime remaining, 11 classes	93 to 97% accuracy, public data
Li et al. 2024, patent CN118310995A	100 kiwifruit, point NIR spectra	Days until a firmness threshold	RMSEP 1.64 days on an 8-day shelf life
Brdar et al. 2021	Tomatoes, hyperspectral at harvest	Fungal infection severity 4 days later	r = 0.92 per tomato

3. Research question and contributions

The central question is how accurately a single VNIR hyperspectral scan of an apple kept at room temperature can predict the number of days until that apple spoils.

This study extends per-fruit spectral prediction to apples, with decay rather than ripeness as the endpoint. Treating the problem as survival analysis with right-censoring lets apples still sound at the end of data collection count as lower bounds on lifetime. Comparing full hyperspectral data with RGB images and 6 to 10 selected wavelengths shows how much information cheaper sensors retain. The project will also release what is, to my knowledge, the first open dataset of daily hyperspectral images of apples, with event times, censoring flags and environmental logs.

The study uses three or four cultivars that spoil at different rates, drawn from McIntosh, Gala, Golden Delicious and Granny Smith and bought from several grocery stores. Cold storage, controlled-atmosphere storage, other fruits, detection of internal rot and consumer devices are out of scope.

4. Experimental setup

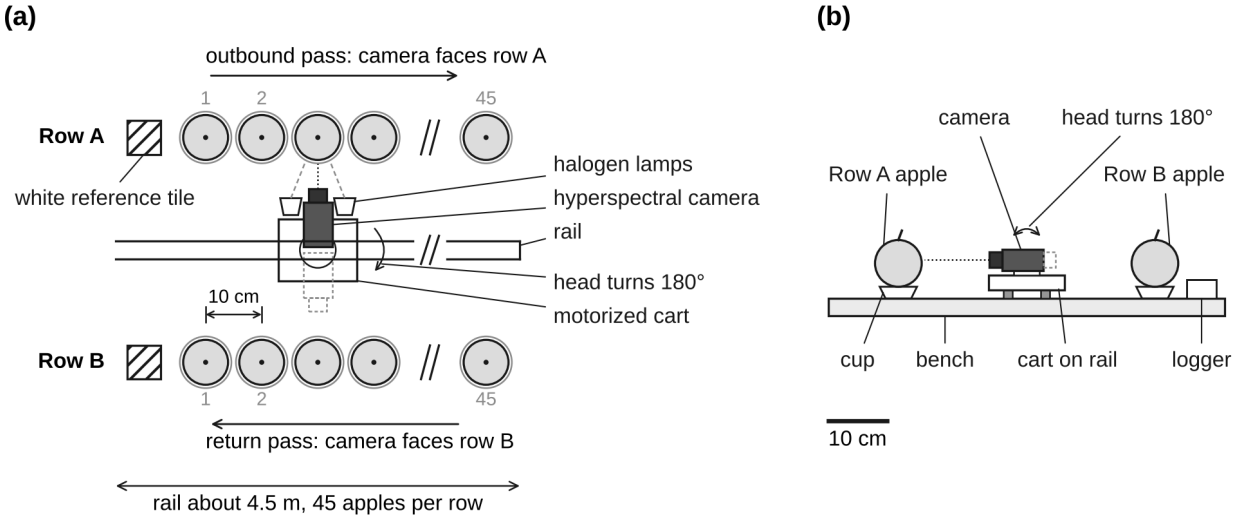


Figure 1. Schematic of the proposed rig. (a) Top view, with one row of apples on each side of a single rail. The camera head faces row A on the outbound pass and row B on the return pass. (b) End view across the rail. Apples are drawn at approximate scale, and the rows are broken to fit.

The apples sit in small cups in two straight rows on a bench, about 45 per row and 10 cm apart center to center, so that neighboring apples do not touch (Figure 1). Each cup holds its apple in a fixed orientation, with the side facing the camera marked. A single rail about 4.5 m long runs midway between the rows, keeping every apple at the same distance from the camera and in focus.

A motorized cart carries the hyperspectral camera (400 to 1000 nm) and a halogen lamp on each side of the lens, all mounted on a head that turns 180 degrees to face either row. Halogen is used because white LEDs emit little light beyond about 700 nm and would leave the near-infrared half of the camera's range unlit. Since the lamps travel with the camera, every apple is lit from the same angle and distance. The lamps are switched on a few minutes before each pass so their output can stabilize, and off afterward to limit heating of the fruit. Each pass also records a white reference tile at the home end of the row and a dark frame with the lamps off, which are used to convert the images to calibrated reflectance. If the camera is a line-scan model, the cart's steady motion forms the image. A camera that scans internally would instead need the cart to stop at each apple.

The room is kept dark and air-conditioned, and a logger records temperature and humidity throughout. A laptop controls the cart, triggers the camera and stores each pass. Depending on camera resolution, the full run will produce a few hundred gigabytes to a few terabytes of data.

5. Protocol

5.1 Daily routine

The cart runs automatically at the same time each day, imaging row A on the outbound pass and row B on the return pass. After the run, I inspect every apple by eye, using a small mirror for the underside, to catch rot the camera cannot see. Apples are not handled after placement, to avoid bruising. When an expiry is confirmed, the apple is removed, cut open to note whether decay began at the core or the skin, and replaced with one from the current purchase batch. Replacement apples are refrigerated until needed, and each new apple's ID, cultivar, store, purchase date and start date are logged.

5.2 Expiry definition

An apple is recorded as expired on the first day it shows rot (a decay lesion or mold patch at least 5 mm across) or shrivel (visible wrinkling of the skin). The sign must persist at the next day's check, and the event is dated to the day it first appeared. The cause, rot or shrivel, is recorded along with the side of the apple where the damage started, so the two can be analyzed as competing risks. Apples that show neither by the end of data collection are right-censored. A second rater independently scores a sample of the same apples, and agreement is measured with Cohen's kappa.

6. Analysis

6.1 Modeling

Each apple has an event day T, counted from the day it was placed on the rig to the day its expiry was first observed. An image taken on day t becomes a training example with target T minus t, and images of censored apples are labeled with a lower bound on the days remaining. For each image the model outputs a survival curve, summarized as a median number of days remaining with an interval. The interval is needed because apples with similar spectra on the same day can spoil several days apart.

The simplest models are Cox and discrete-time hazard models on mean spectra and band indices. Next are deep survival models on the full image, with temperature, humidity and cultivar as covariates. A second version of these starts from self-supervised pretraining on all training images, including those of censored apples. A band-selection analysis on the training data identifies the 6 to 10 most informative wavelengths, which could guide the design of a low-cost multispectral sensor. Data are split by purchase batch, with about one batch in four held out as the test set and one more used for validation, so all images of a given apple fall in the same split. Images of the same apple are strongly correlated, so the effective sample size is set by the number of expiries. Roughly 230 to 320 apples are expected to expire, about three quarters of them in training batches, and the common guideline of about 10 events per predictor (Peduzzi et

al. 1995) limits the main models to a spectrum compressed to about 20 features. Deep models on the full image are treated as exploratory.

6.2 Evaluation

The closest published benchmark is a root-mean-square error of 1.13 to 1.20 days for avocado ripening (Han et al. 2023). Because apples last several times longer than avocados, error is also reported as a fraction of each apple's lifetime. The survival models are evaluated by concordance index, integrated Brier score, mean absolute error in days on apples that expired, and the empirical coverage of the 80% prediction intervals.

The survival models are compared with two baselines. A Kaplan-Meier estimate for each cultivar, equivalent to a single date per batch, shows how much the images add, and partial least squares regression on the mean spectrum, fit only to apples that expired, tests whether the survival framing improves accuracy. The models are also rerun on RGB images rendered from the hyperspectral data and on the selected bands to see how much accuracy cheaper sensors keep. RipeTrack (Waseem et al. 2026) already predicts the remaining lifetime of other fruits from phone RGB and NIR images, and this comparison indicates whether a phone-based approach could work for apples.

7. Plan and timeline

Table 2. Project phases.

Phase	Length	Activities	Outputs
1. Commissioning	First week of operation	The first 90 apples go on the rig while lighting, calibration and scan timing are checked. Days with rig faults are flagged and kept	A working rig, the expiry rule tested on real apples, and the first week of the main dataset
2. Continuous operation	About 12 weeks	Expired apples are replaced from weekly purchase batches (three or four cultivars, several stores), so every cup stays occupied	About 300 to 400 apples with daily image series and event times
3. Analysis and release	Overlaps the final weeks	Modeling, evaluation and write-up	Results, public dataset and code

Data collection starts as soon as a camera is available, beginning with a one-week commissioning period (Table 2). Golden Delicious lasts about four weeks at room temperature (Kassebi et al. 2023), softer cultivars less and Granny Smith longer. Assuming an average of three to four weeks per apple, 12 weeks of operation puts roughly 300 to 400 apples through the rig, counting those still on it at the end.

8. Resources requested

Apart from the hyperspectral camera, the items in Table 3 are inexpensive.

Table 3. Equipment and space.

Item	Purpose	Source
VNIR hyperspectral camera, 400 to 1000 nm	Imaging	Borrowed, or used in a partner lab
One motorized cart with a rotating camera head, and a rail of about 4.5 m	Moves the camera along both rows	Built from off-the-shelf linear-rail parts
Two halogen lamps and a white reference tile	Broadband lighting and calibration	Purchased
About 5 m of bench space and apple cups	Holds the apples in fixed positions	Purchased
Temperature and humidity logger	Environmental record	Purchased
Storage drive of a few terabytes	Image data	Purchased
About 300 to 400 apples	Samples for commissioning and continuous operation	Purchased in weekly batches
A dark, air-conditioned room	Storage and imaging during operation	Lab or departmental space, for the length of the study

9. Risks and limitations

Apples take weeks to spoil, so a rig fault discovered late would cost weeks of data for every apple on the rig. The commissioning week should expose such problems early. Apples can also rot from the core without surface symptoms, limiting the accuracy of any image-based model. Every expired apple, and every apple still on the rig at the end, is cut open to measure how often this happens.

At roughly 300 to 400 apples, the dataset is comparable in size to the avocado studies in Table 1 (80 to 476 fruit) but small for deep models. The camera sees only one side of each apple. Decay that starts on the far side gives the model no warning, although the daily inspection still records it. Judging when rot becomes visible is partly subjective. The next-day rule reduces this error, and the second rater measures it. Store-bought apples come with unknown harvest dates, storage histories and coatings, all of which affect shelf life. Because every apple shares one room, ethylene and mold spores from spoiling apples may hasten their neighbors. A research-grade hyperspectral camera costs tens of thousands of dollars, so the project depends on borrowing one or working with a lab that has one. The literature search covered English-language papers and patents but not Chinese-language databases such as CNKI, where relevant work may exist.

10. Deliverables

The project will release the full image dataset with per-apple event times, censoring flags, cultivar labels and environmental logs, together with the calibration, modeling and evaluation code and the trained models. A written report will present the results, including the most informative wavelengths and how closely the cheaper sensors match full hyperspectral data.

References

Apeel Sciences unveils technology that reveals ripeness with a simple scan. The Packer, 2022. https://www.thepacker.com/news/packer-tech/apeel-sciences-unveils-technology-reveals-ripeness-simple-scan.

Brdar et al. (forecasting tomato fungal infection from harvest-day hyperspectral images). Scientific Reports, 2021. https://doi.org/10.1038/s41598-021-02302-2.

Davur, Kämper, Khoshelham, Trueman, Hosseini Bai. Estimating the ripeness of Hass avocado fruit using deep learning with hyperspectral imaging. Horticulturae 9(5):599, 2023. https://www.mdpi.com/2311-7524/9/5/599.

Detka, Purdy, Melton, Daugovish, Greer, Martin. A plant-level survival modeling framework for spatiotemporal strawberry canopy decline using UAV multispectral time series. Drones 10(4):235, 2026. https://doi.org/10.3390/drones10040235.

Han, Hosseini Bai, Trueman, Khoshelham, Kämper. Predicting the ripening time of 'Hass' and 'Shepard' avocado fruit by hyperspectral imaging. Precision Agriculture 24(5):1889-1905, 2023. https://doi.org/10.1007/s11119-023-10022-y.

Hitachi Digital Services and OneThird, shelf-life scanner announcement, 22 Jan 2024. https://www.hitachi.com/en/press/articles/2024/01/0124c/.

Kassebi, Farkas, Székely, Géczy, Korzenszky. Late shelf life saturation of Golden Delicious apple parameters. Applied Sciences 13(1):159, 2023. https://www.mdpi.com/2076-3417/13/1/159.

Khaled, Ekramirad, Donohue, Villanueva, Adedeji. Non-destructive hyperspectral imaging and machine learning-based predictive models for physicochemical quality attributes of apples during storage as affected by codling moth infestation. Agriculture 13(5):1086, 2023. https://www.mdpi.com/2077-0472/13/5/1086.

Li Hongbo et al. Method and system for non-destructive shelf-life detection of ready-to-eat kiwifruit. Chinese patent application CN118310995A, 2024. https://patents.google.com/patent/CN118310995A/zh.

Ma Huiling, Cao Mengke et al. Fuji apple shelf life prediction with a GAN-improved BP neural network. Transactions of the Chinese Society for Agricultural Machinery, 2021, no. 11, pp. 367-375.

New OneThird technology enables full crate scanning in seconds. Fruit Today. https://fruittoday.com/en/new-onethird-technology-enables-full-crate-scanning-in-seconds/.

Peduzzi, Concato, Feinstein, Holford. Importance of events per independent variable in proportional hazards regression analysis. II. Accuracy and precision of regression estimates. Journal of Clinical Epidemiology 48(12):1503-1510, 1995. https://doi.org/10.1016/0895-4356(95)00048-8.

Stasenko, Shukhratov, Savinov, Shadrin, Somov. Deep learning in precision agriculture: artificially generated VNIR images segmentation for early postharvest decay prediction in apples. Entropy 25(7):987, 2023. https://www.mdpi.com/1099-4300/25/7/987.

Varga, Frank, Zell. Self-supervised pretraining for hyperspectral classification of fruit ripeness. OCM 2023. https://publikationen.uni-tuebingen.de/xmlui/handle/10900/149088?show=full.

Wang Zhihao, Yin Yong, Yu Huichun, Yuan Yunxia, Xue Shuning. Early warning method of apple spoilage based on 2D hyperspectral information representation with pixel mean and variance. Spectroscopy and Spectral Analysis 42(7):2290, 2022. https://www.researching.cn/articles/OJ413c7b1a650930f5.

Wang, Yin, Yu, Yuan. A LIBSVM quality assessment model for apple spoilage during storage based on hyperspectral data. Analytical Methods 16:4765-4774, 2024. https://pubs.rsc.org/en/Content/ArticleLanding/2024/AY/D4AY00678J.

Waseem, Sharma, Hefeeda. RipeTrack: assessing fruit ripeness and remaining lifetime using smartphones. IEEE Transactions on Mobile Computing 25(1):830-845, 2026. Paper: https://www.cs.sfu.ca/~mhefeeda/Papers/tmc26_RipeTrack.pdf; code and data: https://github.com/ShahzaibWaseem/RipeTrack.

Zhao Xin et al. Shelf life identification and quality analysis of Golden Delicious apples based on hyperspectral imaging and near infrared spectroscopy. Science and Technology of Food Industry 46(11):302-312, 2025. https://doi.org/10.13386/j.issn1002-0306.2024080030.